

To aggregate or not to aggregate or how to aggregate?

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Setup

```
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

who_df <- matrix(c(c("1", "Africa"),
                    c("2", "Americas"),
                    c("3", "Eastern Mediterranean"),
                    c("4", "Europe"),
                    c("5", "Southeast Asia"),
                    c("6", "Western Pacific")),
                 ncol = 2, byrow = TRUE) %>%
  data.frame() %>%
  setNames(c("nume", "name"))

who_lst = who_df$name
names(who_lst) = who_df$nume

# and join to data:
dat <- dat %>%
  mutate(who = as.factor(who)) %>%
  left_join(who_df, by = join_by(who == nume)) %>%
  mutate(who_name = ifelse(is.na(name), "Not assigned", name)) %>%
  # put both MIC_nos and MIC_90 in one column
  mutate(mic_mix = case_when(!is.na(micnos_1) ~ micnos_1,
                             !is.na(mic90_1) ~ mic90_1,
                             TRUE ~ NA),
         weight_mic = 1 / mic_mix)
```

Temporal spread

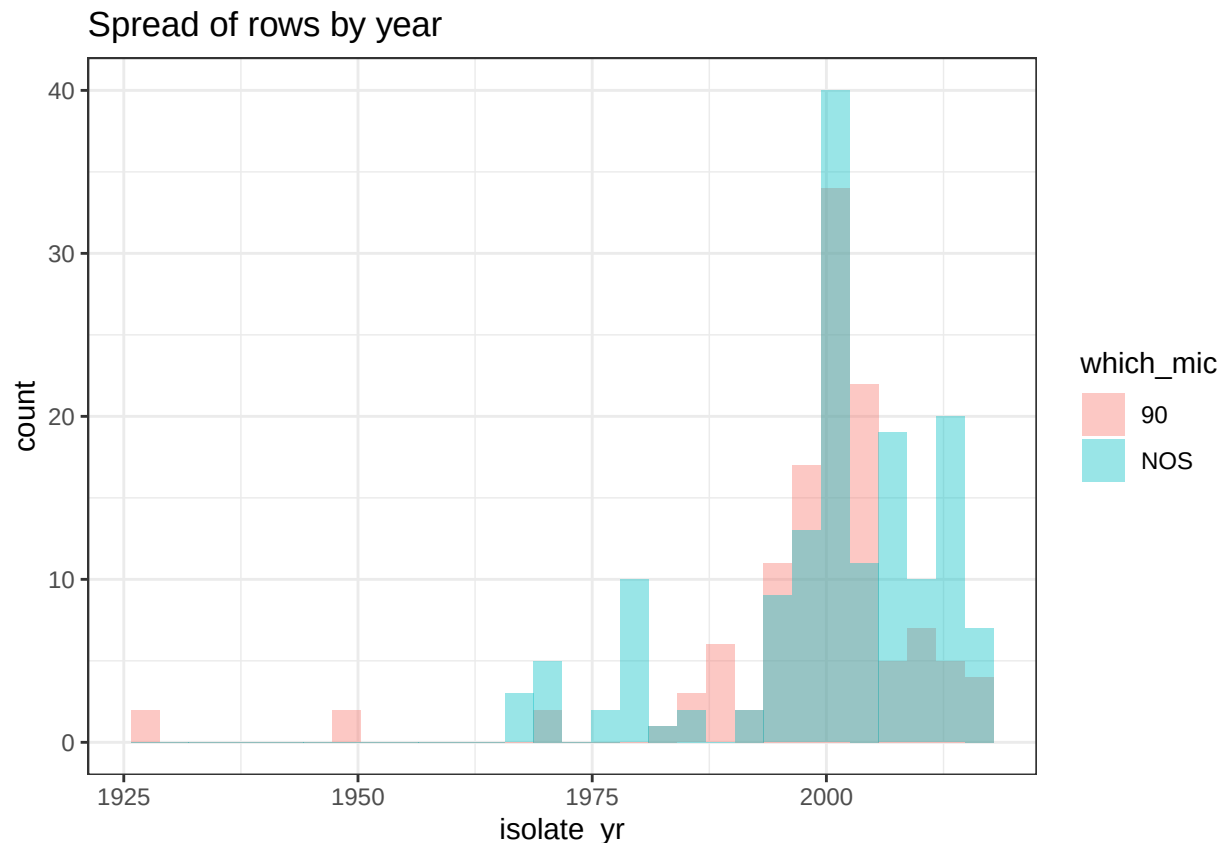
First question - Jeremy has said: “The main limitation being it is my understanding we are taking a spread of values and attributing them all the value representing the 90th percentile, which will, potentially significantly,

overestimate the MIC of the dataset as a whole, rather than comparing the MIC90s reported in the papers over time, as we have a mix of granular MIC and MIC90 data.”

Yes, we would overestimate MIC of the dataset as a whole, but I don’t think this is a huge problem for analysing creep unless there are more/less recent records of either MIC90-only studies or granular studies. Had a look at the temporal spread here:

```
dat <- dat %>%  
  mutate(which_mic = ifelse(!is.na(micnos_1), "NOS", "90"))  
ggplot(dat) +  
  geom_histogram(aes(x = isolate_yr,  
                     group = which_mic,  
                     fill = which_mic),  
                alpha = 0.4, position = "identity") +  
  theme_bw() +  
  labs(title = "Spread of rows by year")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
dat %>% filter(which_mic == "NOS") %>% dplyr::select(isolate_yr) %>% summary()
```

```
## isolate_yr  
## Min. :1967  
## 1st Qu.:1997  
## Median :2000  
## Mean :2000  
## 3rd Qu.:2008  
## Max. :2017
```

```
dat %>% filter(which_mic == "90") %>% dplyr::select(isolate_yr) %>% summary()
```

```
## isolate_yr
## Min.      :1928
## 1st Qu.:1998
## Median :2000
## Mean     :1998
## 3rd Qu.:2003
## Max.     :2017
```

There appear to be more MIC90 studies earlier and more granular studies more recently ... which might cause us to *underestimate* creep if granular records are lower ...

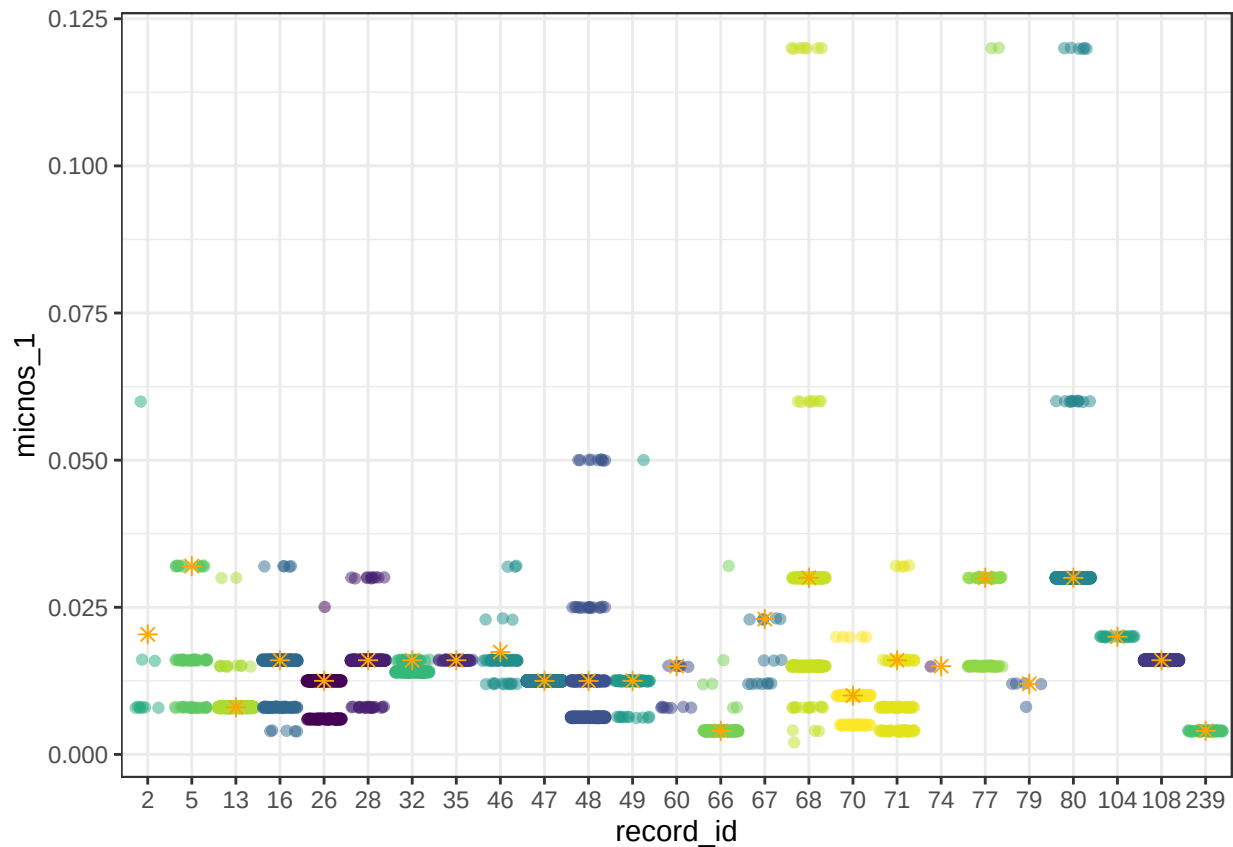
What are we learning from granularity?

Of the studies where we have granular data, what's their spread? Here's a plot of NOSes by study. Asterisks are 90th quantiles for each study ... this is how MIC90 is calculated right? (It kinda loses sense to me when there are less than 10 isolates in a study ... let me know if I've cooked it?)

```
nos <- dat %>%
  filter(which_mic == "NOS") %>%
  # where count > 1, split into multiple rows so that we have multiple dots
  uncount(count) %>%
  mutate(record_id = as.factor(record_id))

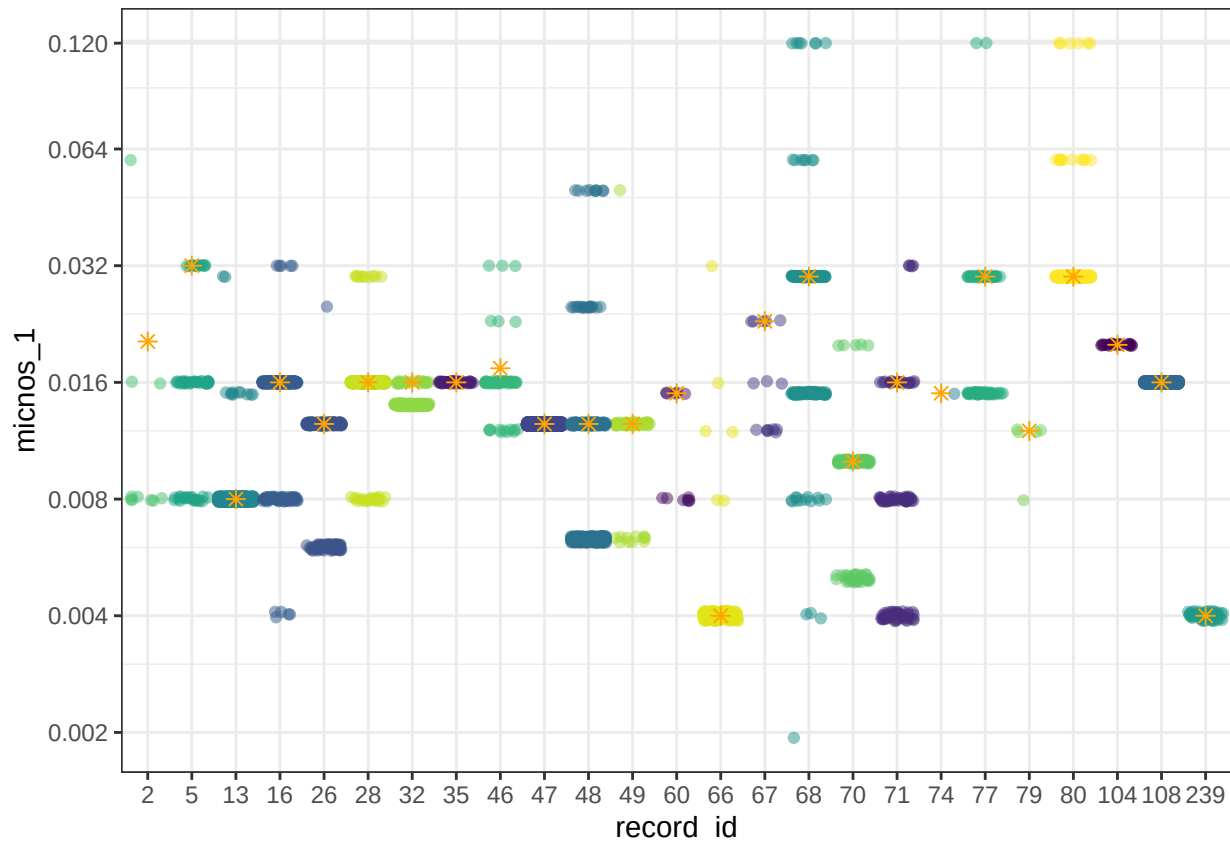
nos_ninetied <- nos %>%
  group_by(record_id) %>%
  summarise(mic90_1 = quantile(micnos_1, probs = 0.9),
            count = n())

ggplot() +
  geom_jitter(data = nos,
             aes(x = record_id, y = micnos_1, col = record_id),
             alpha = 0.5) +
  geom_point(data = nos_ninetied,
            aes(x = record_id, y = mic90_1), pch = 8, col = "orange", size = 2) +
  scale_colour_manual(values = sample(viridis(25))) +
  theme_bw() +
  theme(legend.position = "none")
```



If we consider things on the scale we're doing regression on:

```
ggplot() +
  geom_jitter(data = nos,
             aes(x = record_id, y = micnos_1, col = record_id),
             alpha = 0.5) +
  geom_point(data = nos_ninetied,
            aes(x = record_id, y = mic90_1), pch = 8, col = "orange", size = 2) +
  scale_colour_manual(values = sample(viridis(25))) +
  theme_bw() +
  theme(legend.position = "none") +
  coord_trans(y = "log2") +
  scale_y_continuous(breaks = yticks)
```



Given that lots of these studies involve many isolates, we might be introducing lots of lower MICs that we can't include for the MIC90-only studies ..

Existing model and regression figure

This is the model + visualisation we include in the manuscript. Given tested MICs are doubling, we carry out regression on the log2 scale.

Firstly some run-up:

```
# cobble a palette together ...
pal <- c(rev(viridis(4)),
        pals$loosepal[3],
        iddo_palettes$iddo[1],
        "orange") %>%
  unname()
```

And now for the main event:

```
# apply log to the mics (mix of NOS and 90):
dat <- dat %>%
  mutate(mic_logged = log2(mic_mix))

# fit an unweighted model to determine weights (residual errors)
mod_unweighted = lm(mic_logged ~ isolate_yr, dat)
summary(mod_unweighted)
```

```
##
## Call:
```

```

## lm(formula = mic_logged ~ isolate_yr, data = dat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.99224 -0.40633 -0.07964  0.04772  2.98163
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.9308585   9.0673854  -0.764   0.445
## isolate_yr   0.0004758   0.0045351   0.105   0.917
##
## Residual standard error: 0.9363 on 275 degrees of freedom
## Multiple R-squared:  4.003e-05, Adjusted R-squared:  -0.003596
## F-statistic: 0.01101 on 1 and 275 DF,  p-value: 0.9165

# setting weight to median absolute residuals
wts <- data.frame(isolate_yr = dat$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat <- dat %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr)`
mod_weighted <- lm(mic_logged ~ isolate_yr, dat, weights = wt)
confint(mod_weighted)

##              2.5 %      97.5 %
## (Intercept) -31.570748799  7.90446481
## isolate_yr   -0.006945126  0.01282114

summary(mod_weighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7574 -0.4065 -0.0315  0.0991  3.2507
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -11.833142   10.026073  -1.180   0.239
## isolate_yr   0.002938   0.005020   0.585   0.559
##
## Residual standard error: 0.7866 on 275 degrees of freedom
## Multiple R-squared:  0.001244, Adjusted R-squared:  -0.002388
## F-statistic: 0.3425 on 1 and 275 DF,  p-value: 0.5589

# this is slope on log-scale:
coefficients(mod_weighted)[2]

## isolate_yr
## 0.002938009

```

```
2**(coefficients(mod_weighted)[2])
```

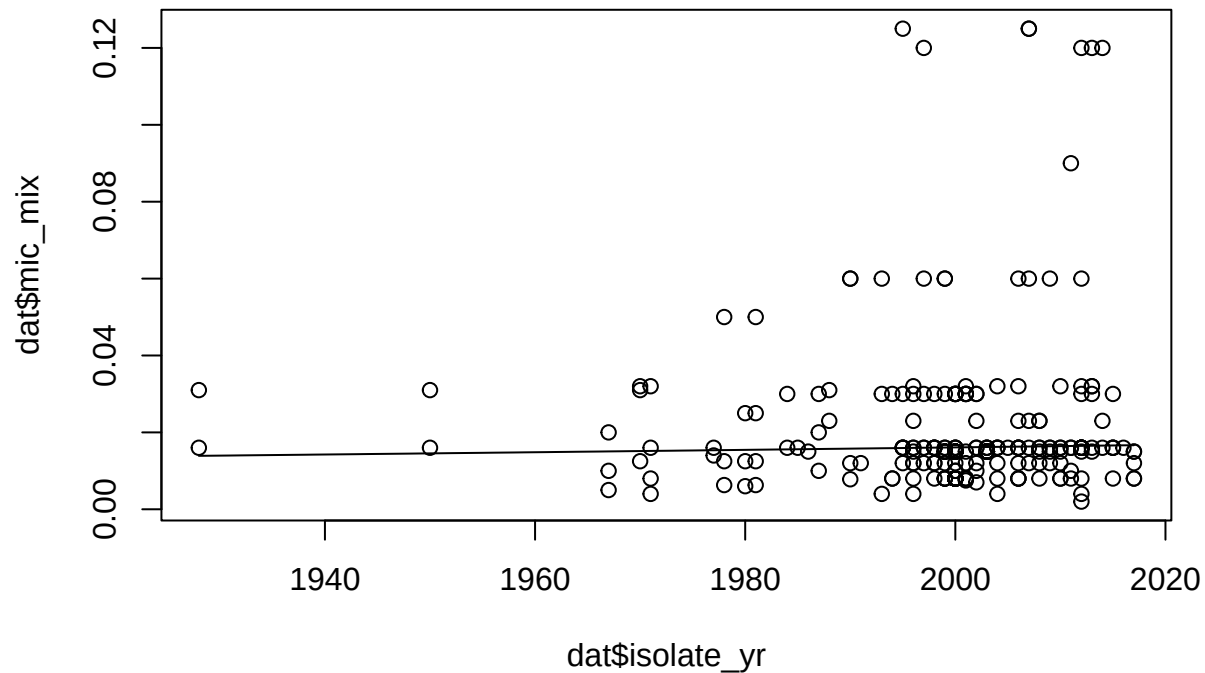
```
## isolate_yr
## 1.002039
```

```
2**confint(mod_weighted)[2,]
```

```
## 2.5 % 97.5 %
## 0.9951976 1.0089265
```

```
# Here's what that looks like unlogged:
```

```
plot(dat$isolate_yr, dat$mic_mix)
yyy = seq(min(dat$isolate_yr), max(dat$isolate_yr))
lines(yyy, 2**predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
# (pretty flat)
```

```
Y_LOWER <- log2(0.0018)
X_LOWER <- 1925
yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

ci <- predict(lm(mic_logged ~ isolate_yr, dat, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat$isolate_yr) + 2)),
              interval = "confidence") %>%
  as.data.frame() %>%
  mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
         isolate_yr = seq(X_LOWER, max(dat$isolate_yr) + 2))

p1 <- ggplot(dat %>%
  filter(!is.na(mic_mix)) %>%
  group_by(isolate_yr, mic_mix, who_name) %>%
  summarise(count = sum(count)) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
```

```

"Americas",
"Eastern Mediterranean",
"Europe",
"Southeast Asia",
"Western Pacific",
"Not assigned")))) %>%

  arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
    alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
    colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = mic_mix,
    col = who_name,
    size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
  theme_bw()

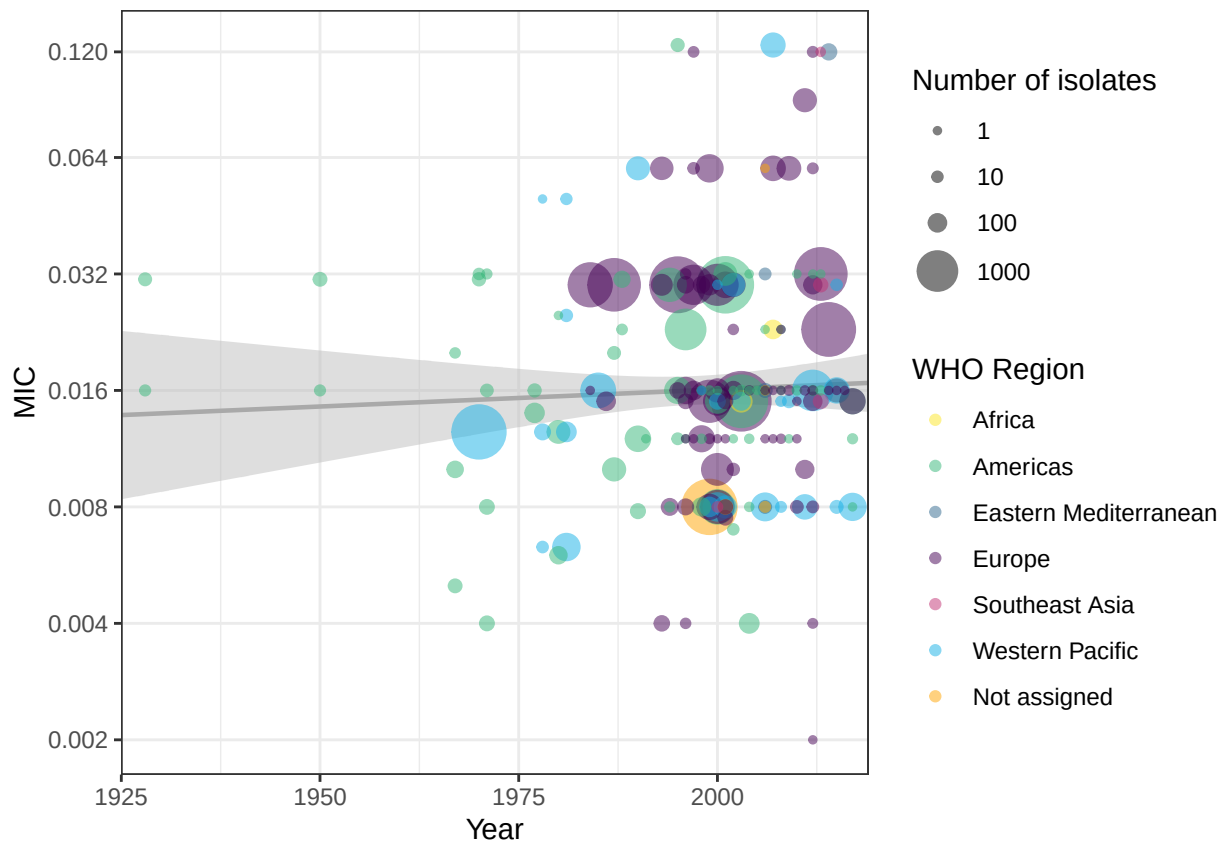
```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, mic_mix, and who_name.
## i Output is grouped by isolate_yr and mic_mix.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, mic_mix, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.

```

p1



The important numbers from above are the slope of our weighted model on the log scale, 0.002938, the slope re-exponentiated, 1.0020385, which is very close to 1, and the CI for the re-exponentiated slope, [0.9951976, 1.0089265], which includes 1.

Model and figure with granular data only

```
# apply log to the granular MICs only:
dat_granular <- dat %>%
  filter(!is.na(micnos_1)) %>%
  mutate(mic_logged = log2(micnos_1))

# everything below this point is as in the previous model:

# fit an unweighted model to determine weights
mod_unweighted = lm(mic_logged ~ isolate_yr, dat_granular)
summary(mod_unweighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_granular)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.93059 -0.81362  0.06941  0.20587  3.12251
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```

## (Intercept) -25.646568 12.497452 -2.052 0.0419 *
## isolate_yr 0.009747 0.006248 1.560 0.1208
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8958 on 152 degrees of freedom
## Multiple R-squared: 0.01576, Adjusted R-squared: 0.009283
## F-statistic: 2.434 on 1 and 152 DF, p-value: 0.1208

wts <- data.frame(isolate_yr = dat_granular$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_granular <- dat_granular %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr, wt)`
mod_weighted <- lm(mic_logged ~ isolate_yr, dat_granular, weights = wt)
confint(mod_weighted)

##                2.5 %      97.5 %
## (Intercept) -48.688696874 0.97778378
## isolate_yr -0.003619368 0.02124957

summary(mod_weighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_granular, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3285 -0.4844  0.0542  0.1596  2.9912
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -23.855457  12.569376  -1.898   0.0596 .
## isolate_yr   0.008815   0.006294   1.401   0.1634
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.704 on 152 degrees of freedom
## Multiple R-squared: 0.01274, Adjusted R-squared: 0.006247
## F-statistic: 1.962 on 1 and 152 DF, p-value: 0.1634
# this is slope on log-scale:
coefficients(mod_weighted)[2]

## isolate_yr
## 0.0088151

2*(coefficients(mod_weighted)[2])

## isolate_yr
## 1.006129

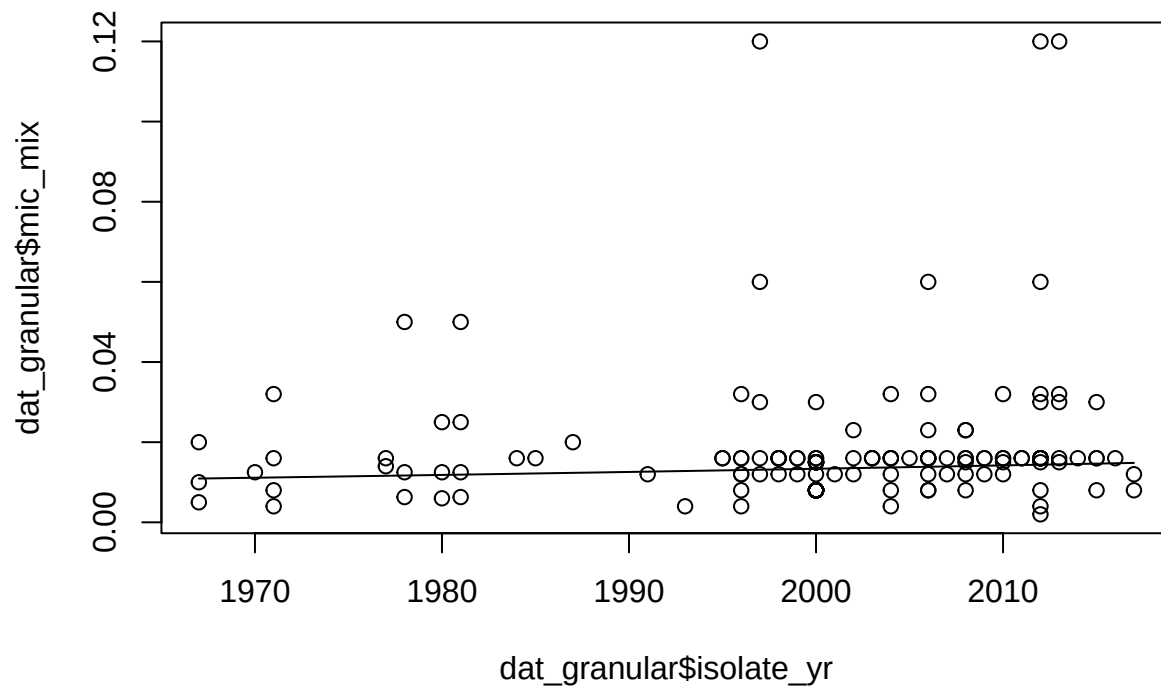
```

```
2**confint(mod_weighted)[2,]
```

```
##      2.5 %      97.5 %  
## 0.9974944 1.0148381
```

```
# Here's what that looks like unlogged:
```

```
plot(dat_granular$isolate_yr, dat_granular$mic_mix)  
yyy = seq(min(dat_granular$isolate_yr), max(dat_granular$isolate_yr))  
lines(yyy, 2**predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
# (pretty flat)
```

```
# Y_LOWER <- log2(0.0018)
```

```
# X_LOWER <- 1962
```

```
# yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
```

```
ci <- predict(lm(mic_logged ~ isolate_yr, dat_granular, weights = wt),  
              newdata = data.frame(isolate_yr = seq(X_LOWER,  
                                                    max(dat_granular$isolate_yr) + 2)),  
              interval = "confidence") %>%  
as.data.frame() %>%  
mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),  
       isolate_yr = seq(X_LOWER, max(dat_granular$isolate_yr) + 2))
```

```
p2 <- ggplot(dat_granular %>%  
             group_by(isolate_yr, micnos_1, who_name) %>%  
             summarise(count = sum(count)) %>%  
             mutate(who_name = factor(who_name, levels = c("Africa",  
                                                            "Americas",  
                                                            "Eastern Mediterranean",
```

```

"Europe",
"Southeast Asia",
"Western Pacific",
"Not assigned")))) %>%

  arrange(desc(count))) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
            alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = micnos_1,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
  theme_bw()

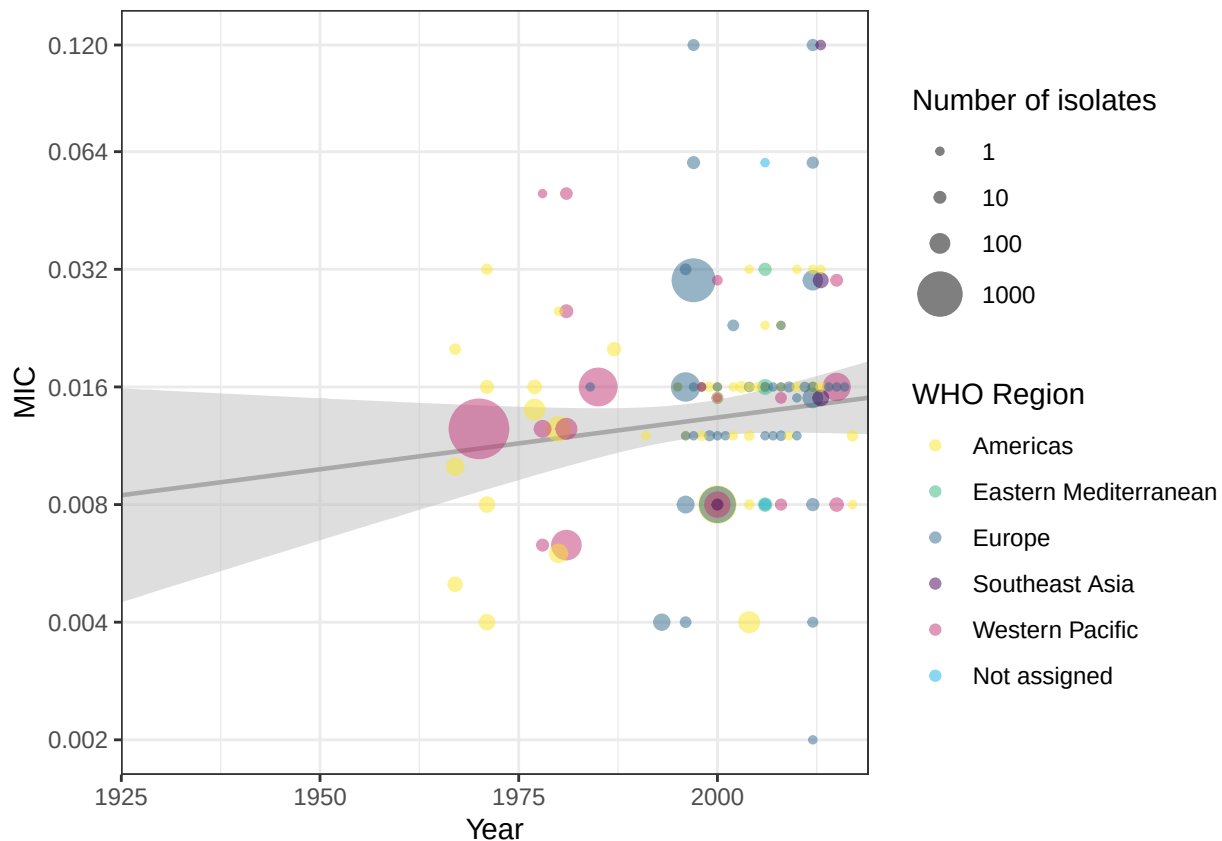
```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.

```

p2



As in the existing model:

The important numbers from above are the slope of our weighted model on the log scale, 0.0088151, the slope re-exponentiated, 1.0061289, which is very close to 1, and the CI for the re-exponentiated slope, [0.9974944, 1.0148381], which includes 1.

Model and figure with MIC90s only

I feel like we started a conversation about aggregating granular data to MIC 90s at some point ... anyway that's what I'm going to try next.

```
# apply log to the granular MICs only:
dat_aggregated <- dat %>%
  filter(!is.na(mic90_1)) %>%
  bind_rows(dat %>%
    filter(is.na(mic90_1)) %>%
    # where count > 1, separate out into individual rows
    uncount(count) %>%
    # have included who_name during grouping, so we effectively end up
    # with more studies than in the scatter plot I did above ...
    group_by(record_id, isolate_yr, who_name) %>%
    summarise(mic90_1 = quantile(micnos_1, probs = 0.9),
              count = n())) %>%
  mutate(mic_logged = log2(mic90_1))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by record_id, isolate_yr, and who_name.
## i Output is grouped by record_id and isolate_yr.
## i Use `summarise(groups = "drop_last")` to silence this message.
```

```
## i Use `summarise(.by = c(record_id, isolate_yr, who_name))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.
```

```
# everything below this point is as in the previous model:
```

```
# fit an unweighted model to determine weights
```

```
mod_unweighted = lm(mic_logged ~ isolate_yr, dat_aggregated)
summary(mod_unweighted)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_aggregated)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1078 -0.4838 -0.1749  0.4207  2.8808
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.618335   9.558321  -0.274   0.784
## isolate_yr  -0.001626   0.004781  -0.340   0.734
##
## Residual standard error: 0.8385 on 187 degrees of freedom
## Multiple R-squared:  0.0006177, Adjusted R-squared:  -0.004727
## F-statistic: 0.1156 on 1 and 187 DF,  p-value: 0.7343
```

```
wt$ <- data.frame(isolate_yr = dat_aggregated$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_aggregated <- dat_aggregated %>%
  left_join(wt$)
```

```
## Joining with `by = join_by(isolate_yr, wt)`
mod_weighted <- lm(mic_logged ~ isolate_yr, dat_aggregated, weights = wt)
confint(mod_weighted)
```

```
##              2.5 %      97.5 %
## (Intercept) -28.41170512 29.24445450
## isolate_yr  -0.01744772  0.01142278
```

```
summary(mod_weighted)
```

```
##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_aggregated,
##     weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7274 -0.4320 -0.1257  0.3360  2.9109
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.416375  14.561374   0.029   0.977
```

```
## isolate_yr -0.003012  0.007291 -0.413    0.680
##
## Residual standard error: 0.7999 on 121 degrees of freedom
## (66 observations deleted due to missingness)
## Multiple R-squared:  0.001409, Adjusted R-squared:  -0.006844
## F-statistic: 0.1707 on 1 and 121 DF, p-value: 0.6802
```

```
# this is slope on log-scale:
coefficients(mod_weighted)[2]
```

```
## isolate_yr
## -0.003012469
```

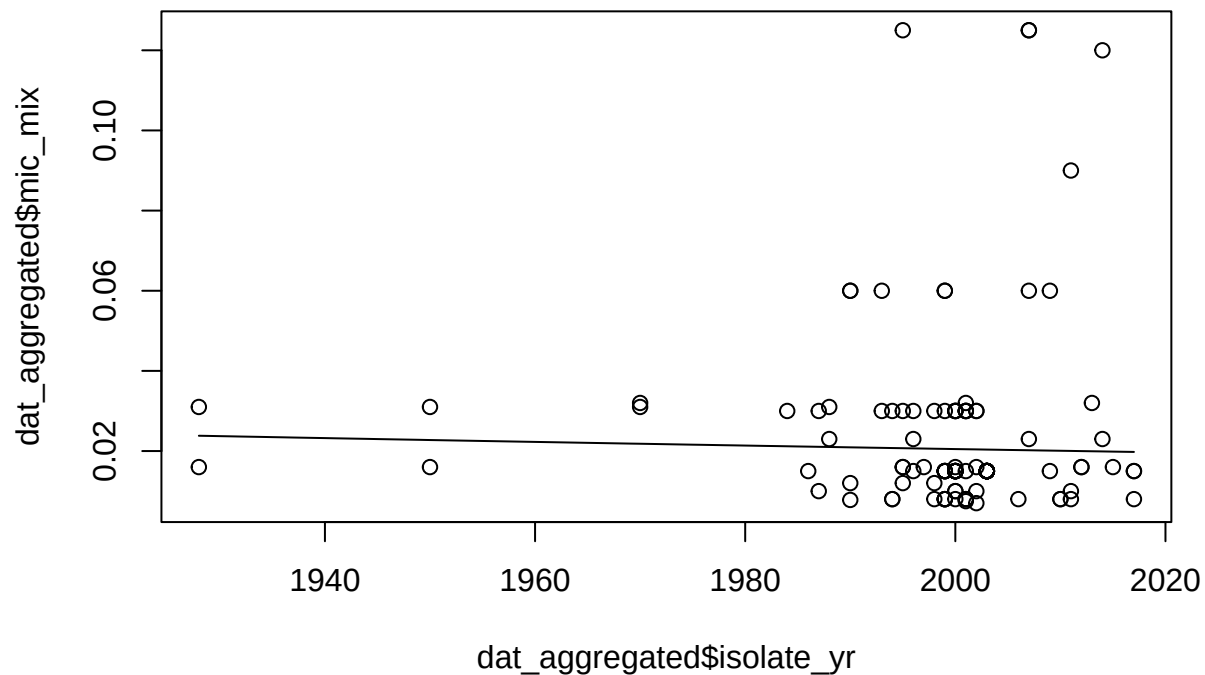
```
2*(coefficients(mod_weighted)[2])
```

```
## isolate_yr
## 0.9979141
```

```
2*confint(mod_weighted)[2,]
```

```
## 2.5 % 97.5 %
## 0.987979 1.007949
```

```
# Here's what that looks like unlogged:
plot(dat_aggregated$isolate_yr, dat_aggregated$mic_mix)
yyy = seq(min(dat_aggregated$isolate_yr), max(dat_aggregated$isolate_yr))
lines(yyy, 2*predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
# (pretty flat)

# Y_LOWER <- log2(0.0018)
# X_LOWER <- 1962
# yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)

ci <- predict(lm(mic_logged ~ isolate_yr, dat_aggregated, weights = wt),
```

```

newdata = data.frame(isolate_yr = seq(X_LOWER,
                                     max(dat_aggregated(isolate_yr) + 2)),
                     interval = "confidence") %>%
as.data.frame() %>%
mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
       isolate_yr = seq(X_LOWER, max(dat_aggregated(isolate_yr) + 2))

p2 <- ggplot(dat_aggregated %>%
             group_by(isolate_yr, mic90_1, who_name) %>%
             summarise(count = sum(count)) %>%
             mutate(who_name = factor(who_name, levels = c("Africa",
                                                           "Americas",
                                                           "Eastern Mediterranean",
                                                           "Europe",
                                                           "Southeast Asia",
                                                           "Western Pacific",
                                                           "Not assigned")))) %>%
             arrange(desc(count))) +
geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
           alpha = 0.5, fill = "grey", colour = NA) +
geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
          colour = "darkgrey", linewidth = 0.8) +
geom_point(aes(x = isolate_yr, y = mic90_1,
               col = who_name,
               size = count), alpha = 0.5) +
scale_color_discrete(name = "WHO Region", type = pal) +
scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
labs(x = "Year", ylab = "MIC") +
coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
theme_bw()

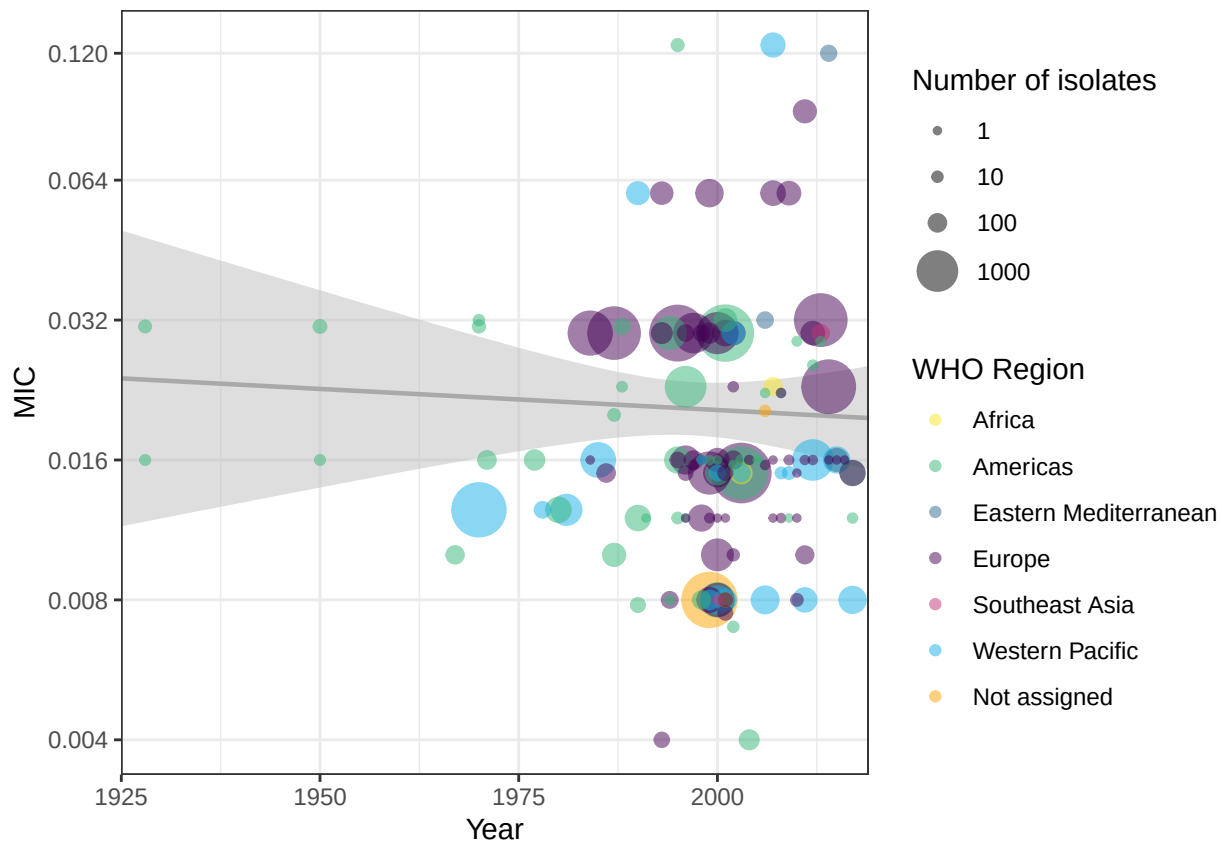
```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, mic90_1, and who_name.
## i Output is grouped by isolate_yr and mic90_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, mic90_1, who_name))` for per-operation
## grouping (`?dplyr::dplyr_by`) instead.

```

p2



As in the existing model:

The important numbers from above are the slope of our weighted model on the log scale, -0.0030125 , the slope re-exponentiated, 0.9979141 , which is very close to 1, and the CI for the re-exponentiated slope, $[0.987979, 1.0079491]$, which includes 1.

Which is not what I would have expected at all: when we include the granular studies as aggregate MIC90s, we get to a model with a lower slope creep than when we include the granular values ...

Bonus: Uncounted model and figure with granular data only

Question: we're not including numbers of isolates as part of our model ... what happens if we do? Here, I've disaggregated the granular dataset so that we have a row for each isolate.

```
# apply log to the granular MICs only:
dat_granular <- dat %>%
  filter(!is.na(micnos_1)) %>%
  mutate(mic_logged = log2(micnos_1)) %>%
  uncount(count)

# everything below this point is as in the previous model:

# fit an unweighted model to determine weights
mod_unweighted = lm(mic_logged ~ isolate_yr, dat_granular)
summary(mod_unweighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_granular)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8081 -0.7358  0.0887  0.2883  3.1891
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.828e+01  1.131e+00  -16.17  <2e-16 ***
## isolate_yr   6.024e-03  5.687e-04   10.59  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7211 on 7649 degrees of freedom
## Multiple R-squared:  0.01446,    Adjusted R-squared:  0.01433
## F-statistic: 112.2 on 1 and 7649 DF,  p-value: < 2.2e-16

wts <- data.frame(isolate_yr = dat_granular$isolate_yr,
                  resid = mod_unweighted$residuals) %>%
  group_by(isolate_yr) %>%
  summarise(wt = median(abs(resid)))

dat_granular <- dat_granular %>%
  left_join(wts)

## Joining with `by = join_by(isolate_yr, wt)`
mod_weighted <- lm(mic_logged ~ isolate_yr, dat_granular, weights = wt)
confint(mod_weighted)

##              2.5 %          97.5 %
## (Intercept) -7.93954290 -3.4023419014
## isolate_yr  -0.00153302  0.0007527708

summary(mod_weighted)

##
## Call:
## lm(formula = mic_logged ~ isolate_yr, data = dat_granular, weights = wt)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1045 -0.4937  0.1165  0.1165  3.3300
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -5.6709424  1.1572873  -4.900 9.77e-07 ***
## isolate_yr  -0.0003901  0.0005830  -0.669   0.503
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.598 on 7649 degrees of freedom
## Multiple R-squared:  5.853e-05,    Adjusted R-squared:  -7.22e-05
## F-statistic: 0.4477 on 1 and 7649 DF,  p-value: 0.5034

# this is slope on log-scale:
coefficients(mod_weighted)[2]
```

```
## isolate_yr
## -0.0003901244
```

```
2**(coefficients(mod_weighted)[2])
```

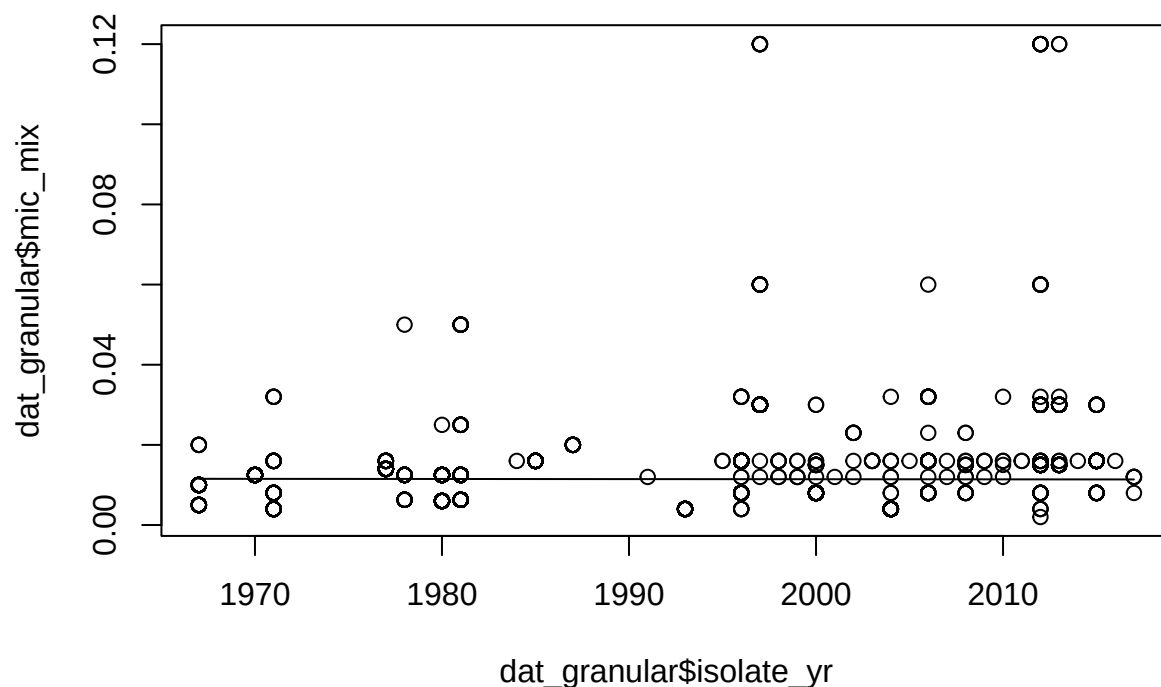
```
## isolate_yr
## 0.9997296
```

```
2**confint(mod_weighted)[2,]
```

```
## 2.5 % 97.5 %
## 0.998938 1.000522
```

```
# Here's what that looks like unlogged:
```

```
plot(dat_granular$isolate_yr, dat_granular$mic_mix)
yyy = seq(min(dat_granular$isolate_yr), max(dat_granular$isolate_yr))
lines(yyy, 2**predict(mod_weighted, data.frame(isolate_yr = yyy)))
```



```
# (pretty flat)
```

```
# Y_LOWER <- log2(0.0018)
```

```
# X_LOWER <- 1962
```

```
# yticks <- c(0, 0.002, 0.004, 0.008, 0.016, 0.032, 0.064, 0.12)
```

```
ci <- predict(lm(mic_logged ~ isolate_yr, dat_granular, weights = wt),
              newdata = data.frame(isolate_yr = seq(X_LOWER,
                                                    max(dat_granular$isolate_yr) + 2)),
              interval = "confidence") %>%
as.data.frame() %>%
mutate(lwr = ifelse(lwr < Y_LOWER, Y_LOWER, lwr),
       isolate_yr = seq(X_LOWER, max(dat_granular$isolate_yr) + 2))
```

```

p2 <- ggplot(dat_granular %>%
  group_by(isolate_yr, micnos_1, who_name) %>%
  summarise(count = n()) %>%
  mutate(who_name = factor(who_name, levels = c("Africa",
                                                "Americas",
                                                "Eastern Mediterranean",
                                                "Europe",
                                                "Southeast Asia",
                                                "Western Pacific",
                                                "Not assigned")))) %>%

  arrange(desc(count)) +
  geom_ribbon(data = ci, aes(ymin = 2**lwr, ymax = 2**upr, x = isolate_yr),
            alpha = 0.5, fill = "grey", colour = NA) +
  geom_line(data = ci, aes(x = isolate_yr, y = 2**fit),
            colour = "darkgrey", linewidth = 0.8) +
  geom_point(aes(x = isolate_yr, y = micnos_1,
                 col = who_name,
                 size = count), alpha = 0.5) +
  scale_color_discrete(name = "WHO Region", type = pal) +
  scale_size_continuous(range = c(1,10), name = "Number of isolates", breaks = c(1, 10, 100, 1000)) +
  scale_y_continuous(breaks = yticks, minor_breaks = NULL, name = "MIC") +
  scale_x_continuous(limits = c(X_LOWER, 2017 + 2), expand = c(0,0)) +
  labs(x = "Year", ylab = "MIC") +
  coord_trans(y = "log2") + #, ylim = c(Y_LOWER, 0.14), expand = FALSE) +
  theme_bw()

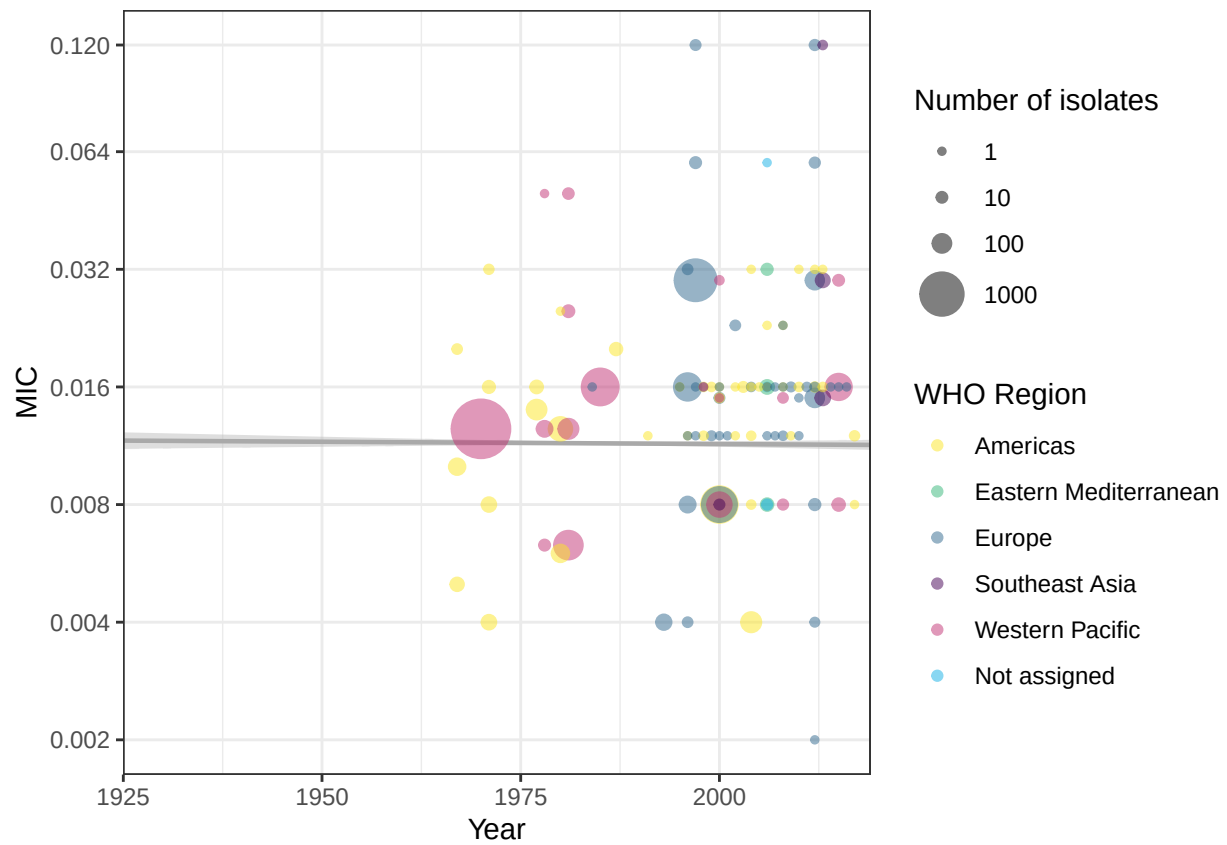
```

```

## `summarise()` has regrouped the output.
## i Summaries were computed grouped by isolate_yr, micnos_1, and who_name.
## i Output is grouped by isolate_yr and micnos_1.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(isolate_yr, micnos_1, who_name))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.

```

p2



We have a very narrow CI, shrunk down by our very large sample size . . . unsure if this is the sort of thing we should show. Is it fair to assume that each of those isolates is independent?